Chapter 6 Concept Checkpoint — Energy Transformations

Grade 4 · Energy Transfer and Transformation · Week 7

Name: _____ Date: _____

Circle or write the best answer.

PART 1 — Core Ideas

1. Energy transformation means energy ___.

a) disappears b) changes form c) stops moving d) becomes nothing

2. Energy transfer means energy ___.

a) changes form b) moves from one object or place to another c) disappears d) turns into light

3. A flashlight changes stored energy to light. This is a ___.

a) transformation b) transfer only c) nothing d) loss of energy

4. Falling dominoes knock over the next domino AND make a click. This is ___.

a) transfer only b) transformation only c) BOTH transfer and transformation d) neither

PART 2 — Energy Chain

5. Fill in the chain for a wind-up toy:

INPUT = stored energy → CHANGE = _____ → OUTPUT = the toy moves

PART 3 — Engineering Design

6. The desired features a good solution must have are called ___.

a) constraints b) criteria c) tests d) materials

7. The limits on a design (materials, cost, time, safety) are called ___.

a) criteria b) problems c) constraints d) signals

8. After a test, engineers use the results to ___.

a) stop forever b) refine the design c) hide the problem d) start over without thinking

PART 4 — Evidence

9. A student's device worked in Test 1 but not in Test 2. What should the team do?

a) give up b) look at the evidence and change ONE thing c) pretend it worked d) throw it away

10. Finish this sentence using the words “evidence” and “supports” (NOT “proves”):

Our observation that the bell rang louder is ___ that ___ the change helped.